

- Web-based materials/chemistry simulation platform

Materials Square Class Curriculum

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[Lab.2] Li diffusion activation barrier in bulk structure

Learning Objectives:

→ Analyze the energy profile of Li diffusion between two adjacent Li sites and the corresponding Li migration path.

Materials: LiFePO_4 Bulk

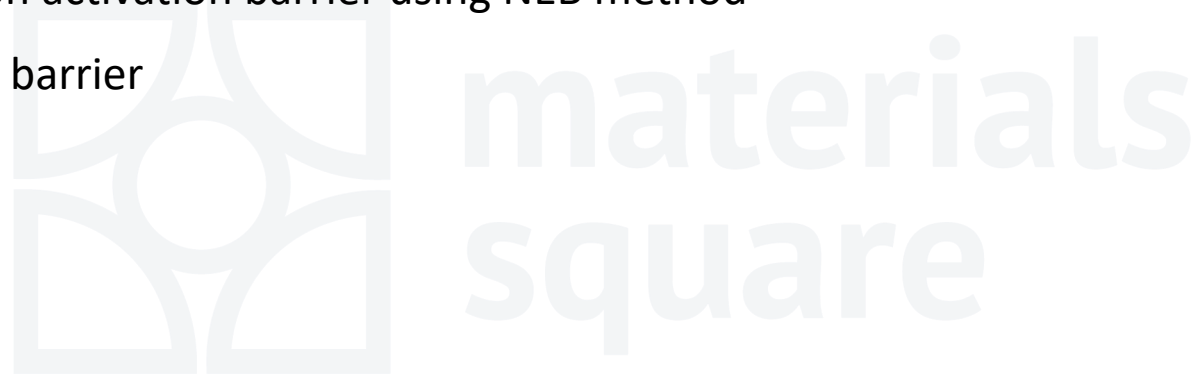


MatSQ Credit: ~ 12 CPU hour (\$ 3)

[Lab.2] Li diffusion activation barrier in bulk structure

LiFePO₄-bulk

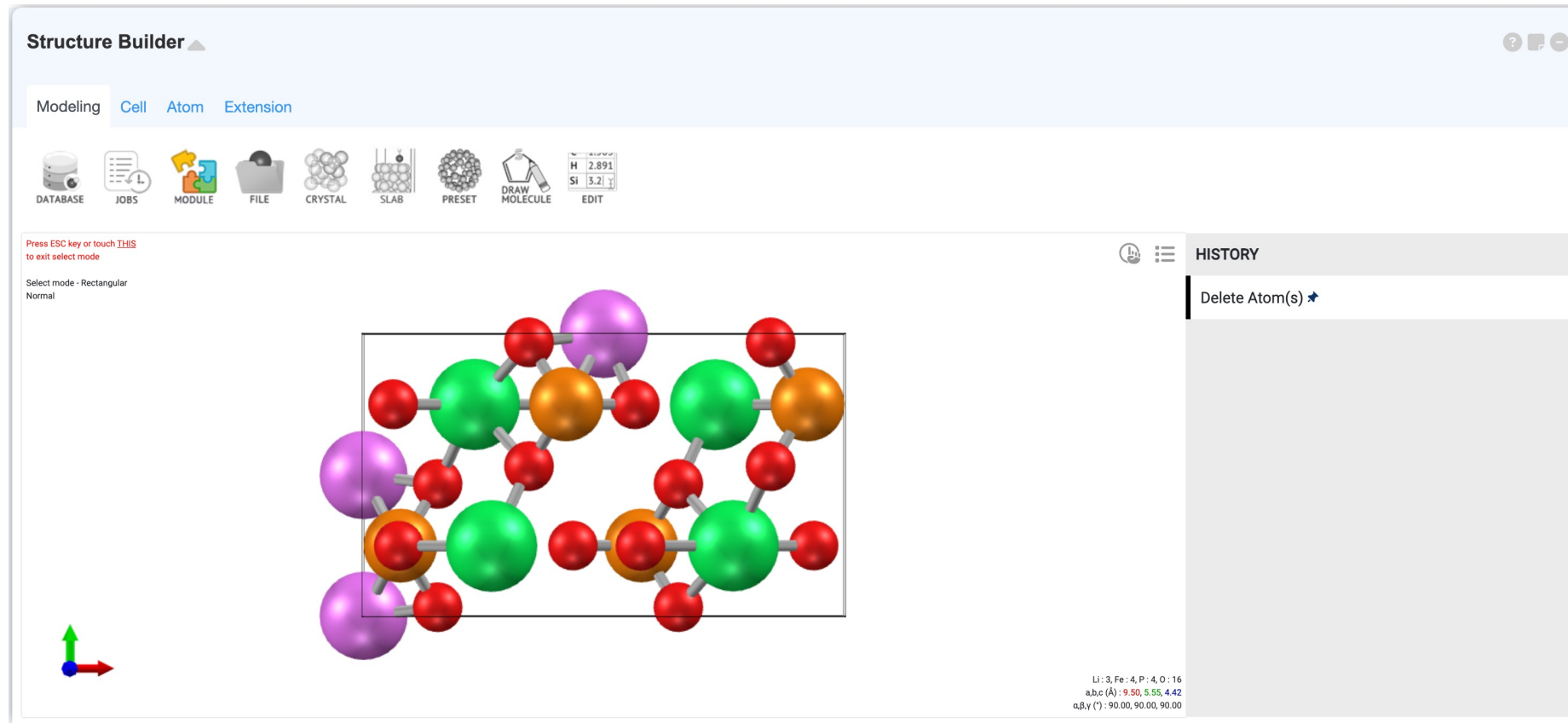
1. Build bulk structure with Li vacancy (initial, final)
2. Calculate Li diffusion activation barrier using NEB method
3. Analyze the energy barrier



MatSQ Credit: ~ 12 CPU hour (\$ 3)

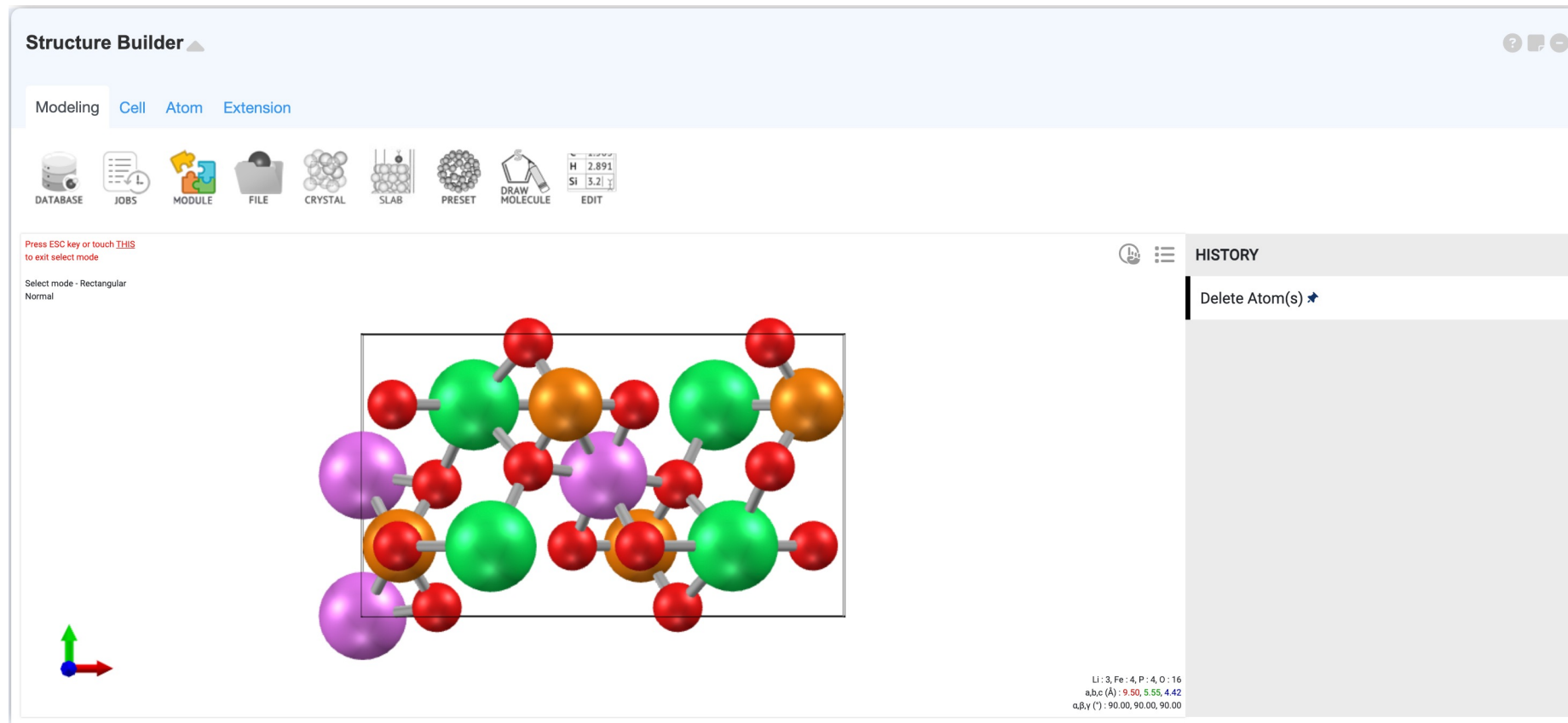
[Lab.2] Li diffusion activation barrier in bulk structure

1. Build bulk structure with Li vacancy (initial, final)



[Lab.2] Li diffusion activation barrier in bulk structure

1. Build bulk structure with Li vacancy (initial, final)



[Lab.2] Li diffusion activation barrier in bulk structure

2. Calculate Li diffusion activation barrier using NEB method

QE NEB

NEB Result

Scripting Option : General Atom info Update Structure Keyword information

&PATH

Relaxation steps1

NEB images9

Optimization methodDefault

ClimbingDefault

Optimization step length (Hartree a.u.)1.D0

Max elastic constant (Hartree a.u.)0.1D0

Min elastic constant (Hartree a.u.)0.1D0

PWSCF Input Script

```
&CONTROL
calculation = 'scf'
outdir = './output/'
prefix = 'VLAB'
pseudo_dir = './'
restart_mode = 'from_scratch'
wf_collect = .TRUE.
/

&SYSTEM
degauss = 0.002
ecutwfc = 30
```

input.neb.x Potential

```
BEGIN
BEGIN_PATH_INPUT
&PATH
string_method = 'neb'
nstep_path = 1
k_min = 0.1
k_max = 0.1
restart_mode = 'from_scratch'
num_of_images = 9
ds = 1.0
/
END_PATH_INPUT
```

Job Submit

Resource : Spot

PPN : 48 ram : 192 GB

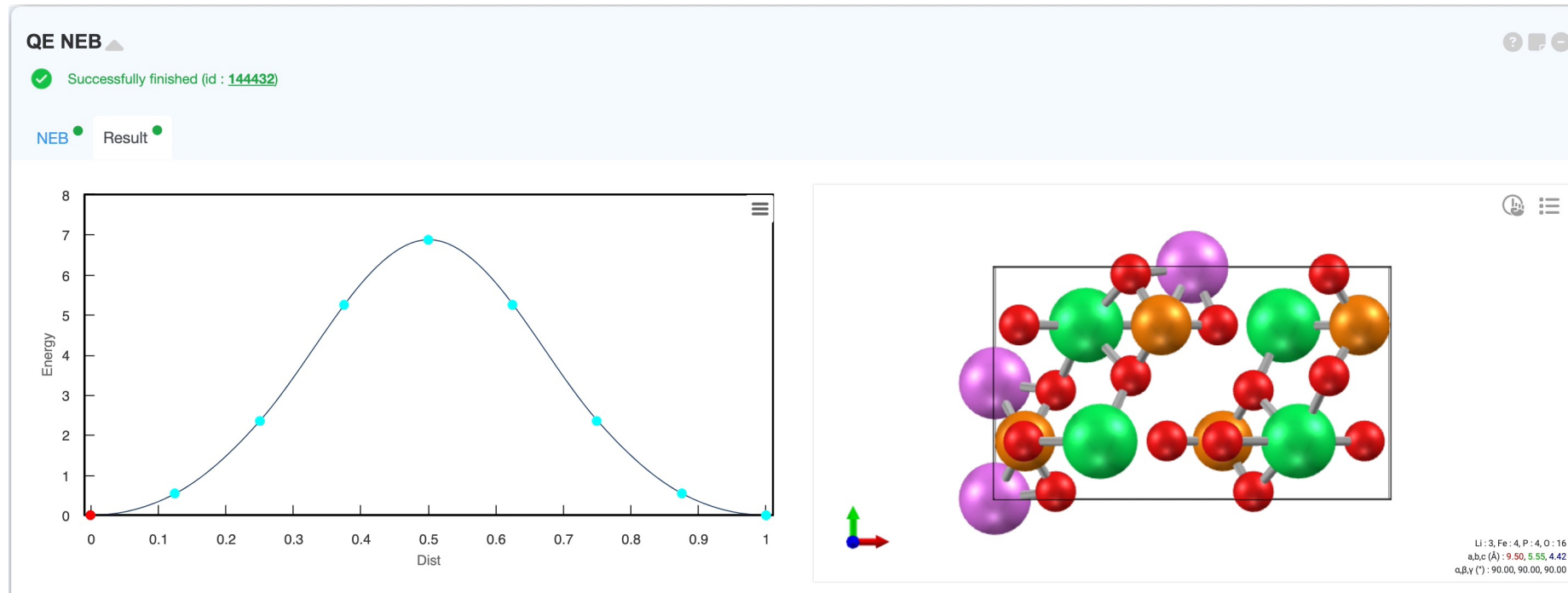
Node type : Computing

Job Name : neb

Version : 6.4

[Lab.2] Li diffusion activation barrier in bulk structure

3. Analyze the energy barrier



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Thank you

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